Practical Task 4.3

(Credit Task)

Submission deadline: Monday, April 22
Discussion deadline: Friday, May 8

Instructions

You have just arrived at your practical class to participate in a face-to-face discussion of your tasks when your worst nightmare comes true. Your university campus is progressively being infected by a mutated version of Cordyceps. It infects humans, controls their behavior, and causes rapid brain infection and physical mutations. The campus has been closed, enclosed with a high wall, and declared a quarantine zone.

It is now a rectangular maze divided into *C* columns and *R* rows. Both rows and columns are numbered starting from 0. Each cell is either empty or labeled with a character:

- '#' for an internal wall,
- 'F' for a known source of Cordyceps infection, and
- '$' for your location on campus.

Cordyceps spreads every hour, occupying new empty cells. It cannot pass through walls but can move to an adjacent empty cell from any cell that is already infected. Two cells are considered adjacent if they share a side.

Similarly, every hour you may either stay in your current location or move one cell to the east, west, north, or south. However, you cannot pass through walls.

While your classmates are working on a plan to help you escape from the restricted zone, you decide to create your own plan to survive as many hours as possible. Your task is therefore straightforward: you must use your laptop to write a program that finds a path in the maze and reports to your classmates the number of hours you can survive — or -1 if you can survive indefinitely.

1. Use the source code attached to this task. Create a new project and import the three files. Your newly built project should compile without errors. Note that the Tester.cs file provides you with a Main method as the starting point of your program and is important for the purpose of debugging and testing. Another file, TestGenerator.cs, serves the Tester class in Tester.cs with hard-coded test instances. Finally, FungusInfestation.cs contains a template for the algorithm that you need to develop and implement as part of the FungusInfestationclass.
2. Explore the existing FungusInfestation class to find the static method named Solve. This method must implement your algorithm. It accepts only a one-dimensional array of strings, with string a new row representing the maze. The first string represents the top of the maze, and the last is its bottom. Each string comprises letters representing the respective cells: '#', '$', 'F', and ',' that stands for an empty space.
3. Consider the following examples:

    1. **{"F..", ".$.", "..."}**

       Returns: 4. Your best move in this case is to go to the bottom right cell. Cordyceps will reach that cell after the fourth hour.

    2. **{".F#...", "F....#", ".F###.", "F.#.$.", "F.#..."}**

       Returns: -1. There is a wall barrier between you and the multiple cells that will be occupied by Cordyceps. It is possible to survive indefinitely.

3. **{"....#.", "$##.#.", ".#..#F", ".F#.#.", "..#..."}**

   Returns: 7

4. **{"...$..", "..#...", "..###.", "..#...", "F.#.F."}**

   Returns: 7 Sometimes it is best not to move the character at all.

5. **{".F....F.", ".#.##.#.", ".#....#.", "F.$##..F", ".#....#.", ".###.##.", ".F....F."}**

   Returns: 4

6. **{"F#....", ".####.", "......", ".####.", "$##..."}**

   Returns: 4

4. Remember that you are free to write your program code within the FungusInfestation.cs as you wish. The only requirement is to meet the expected signature of the Solve method. Therefore, you may add any extra private methods and attributes if necessary. Remember that computer memory is another important resource that you have access to, and its use can reduce algorithm's complexity and simplify its implementation.
5. As you progress with the implementation of your algorithm, you should start using the Tester class in order to test your code for potential logical issues and runtime errors. This (testing) part of the task is as important as coding.
6. Write a report (no more than 1 page) documenting your algorithm. The report must discuss the algorithmic approach you picked to solve the problem and use a pseudocode to clearly explain the core steps of your algorithm. If you see it useful, feel free to complement your explanation with schemes, diagrams, and small numeric examples. The main goal is to clearly communicate the way you solved the problem to someone who knows the basics of problem-solving and may wish to reimplement the algorithm relying on your report.
7. Finally, note that cheating is not allowed; that is, it is unacceptable to simply return the expected answers of a test instance you are given. Such illegal solutions will be marked as failed.

Further Notes

- What should help you to develop an algorithm to this problem is Lecture 4, particularly its "Blob Check" example.
- What may inspire you without giving any answer to this task is watching "The Last of Us".

Submission Instructions and Marking Process

To get your task completed, you must finish the following steps strictly on time.

- Make sure your programs implement the required functionality. They must compile, have no runtime errors, and pass all test cases of the task. Programs causing compilation or runtime errors will not be accepted as a solution. You need to test your programs thoroughly before submission. Think about potential errors where your programs might fail.
- **Submit** the expected code files as a solution to the task via OnTrack submission system.
- Once your solution is accepted by your tutor, you will be invited to **continue its discussion and answer relevant theoretical questions through a face-to-face interview**. Specifically, you will need to meet with the tutor to demonstrate and discuss the solution in one of the dedicated practical sessions (run online via MS Teams for online students and on-campus for students who selected to join classes at Burwood\Geelong). Please, come prepared so that the class time is used efficiently and fairly for all students in it. Be on time with respect to the specified discussion deadline.
  You will also need to **answer all additional questions** that your tutor may ask you. Questions will cover the lecture notes; so, attending (or watching) the lectures should help you with this **compulsory** discussion part. You should start the discussion as soon as possible as if your answers are wrong, you may have to pass another round, still before the deadline. Use available attempts properly.

Note that we will not accept your solution after **the submission deadline** and will not discuss it after **the discussion deadline**. If you fail one of the deadlines, you fail the task, and this reduces the chance to pass the unit. Unless extended for all students, the deadlines are strict to guarantee smooth and on-time work throughout the unit.

Remember that this is your responsibility to keep track of your progress in the unit that includes checking which tasks have been marked as completed in the OnTrack system by your marking tutor, and which are still to be finalised. When grading your achievements at the end of the unit, we will solely rely on the records of the OnTrack system and feedback provided by your tutor about your overall progress and the quality of your solutions.

Appendix A: Expected Printout

This section displays the printout produced by the attached Tester class, specifically by its Main method. It is based on our solution. The printout is provided here to help with testing your code for potential runtime errors and logical mistakes in your algorithm.

```
Attempting test instance 0 with the expected answer 4 for matrix
F..
.$.
...
 :: SUCCESS (Time elapsed 00:00:00.0216413)

Attempting test instance 1 with the expected answer -1 for matrix
.F#...
F....#
.F###.
F.#.$.
F.#...
 :: SUCCESS (Time elapsed 00:00:00.0000560)

Attempting test instance 2 with the expected answer 7 for matrix
....#.
$##.#.
.#..#F
.F#.#.
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..#...
 :: SUCCESS (Time elapsed 00:00:00.0000214)

Attempting test instance 3 with the expected answer 7 for matrix
...$..
..#...
..###.
..#...
F.#.F.
 :: SUCCESS (Time elapsed 00:00:00.0000144)

Attempting test instance 4 with the expected answer 4 for matrix
.F....F.
.#.##.#.
.#....#.
F.$##..F
.#....#.
.###.##.
.F....F.
 :: SUCCESS (Time elapsed 00:00:00.0000561)

Attempting test instance 5 with the expected answer 4 for matrix
F#....
.####.
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.####.
$##...
 :: SUCCESS (Time elapsed 00:00:00.0000210)

Attempting test instance 6 with the expected answer 1274 for matrix
F..................................................
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 :: SUCCESS (Time elapsed 00:00:00.0005987)

Attempting test instance 7 with the expected answer 98 for matrix
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 :: SUCCESS (Time elapsed 00:00:00.0056960)

Attempting test instance 8 with the expected answer 50 for matrix
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 :: SUCCESS (Time elapsed 00:00:00.0010402)
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Attempting test instance 9 with the expected answer 72 for matrix
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 :: SUCCESS (Time elapsed 00:00:00.0029401)

Attempting test instance 10 with the expected answer 47 for matrix
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#.#####.##....###################.#######.####.###
#.#####.###.F.###################.#######.####.###
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#.#####.################..........#######.####.###
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 :: SUCCESS (Time elapsed 00:00:00.0030470)


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Attempting test instance 109 with the expected answer 4 for matrix
F...
#.#$
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 :: SUCCESS (Time elapsed 00:00:00.0000105)


Summary: 110 tests out of 110 passed
Tests passed (0 to 110): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20,
21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45,
46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70,
71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95,
96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109
Tests failed (0 to 110): none
```